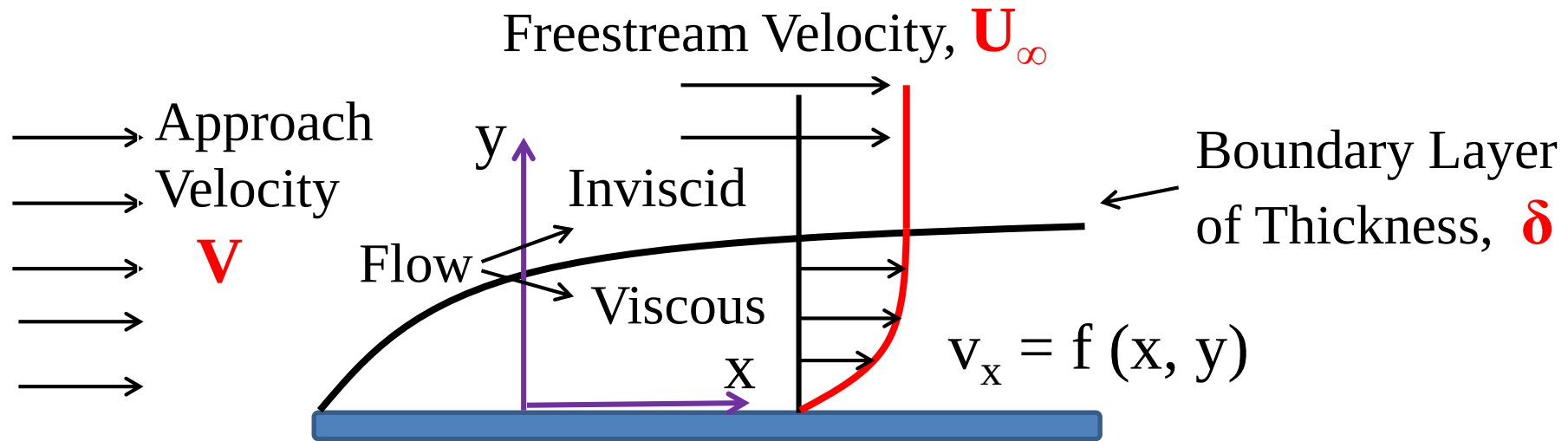


External Incompressible Viscous Flow – Boundary Layer



Flow over a flat plate

$v_x = f(x, y)$ Viscous 2D flow inside BL

$v_x = U_\infty$ Inviscid flow outside BL

$v_x = 0.99U_\infty$
at $y = \delta$

δ - boundary layer thickness

Macroscopic Balance

MACROSCOPIC BALANCE

For § McDonald

FOR A
CV

$$\frac{dN}{dt} \Big|_{\text{sys}} = \frac{\partial}{\partial t} \int_{CV} \eta \rho dV + \int_{CS} \eta \rho \vec{V} \cdot d\vec{A} \quad - (1)$$

$\frac{dN}{dt}$ = TOTAL RATE OF CHANGE OF AN
ARBITRARY EXTENSIVE PROP.
OF THE SYSTEM

N = EXTENSIVE
PROPERTY

$\eta = \frac{N}{m}$, INTENSIVE
PROPERTY

$\frac{\partial}{\partial t} \int_{CV} \eta \rho dV$ = TIME RATE OF CHANGE OF THE EXTENSIVE
PROP. 'N' WITHIN CV

$\int_{CS} \eta \rho \vec{V} \cdot d\vec{A}$ = NET RATE OF EFFLUX OF THE EXTENSIVE
PROPERTY 'N' THROUGH THE CONTROL SURFACES

Macroscopic Balance

NOTES
1.1.1.1.1.1

$$\frac{dN}{dt} |_{\text{SYSTEM}} = \frac{\partial}{\partial t} \int_{CV} \eta \rho dV + \int_{CS} \eta \rho \vec{V} \cdot d\vec{A}$$

MASS

$$N = M, \quad \eta = \frac{N}{M} = 1$$

$$0 = \frac{\partial}{\partial t} \int_{CV} \rho dV + \int_{CS} \rho \vec{V} \cdot d\vec{A}$$

CONSERVATION
OF MASS
CONTINUITY
EQⁿ.

STEADY STATE
INCOMP.

$$\rho \int_{CS} \vec{V} \cdot d\vec{A} = 0 \Rightarrow V_1 A_1 + V_2 A_2 + \dots = 0$$

$$\sum V_i A_i = 0$$

MOMENTUM

$$\eta = \frac{N}{M} = \frac{\vec{P}}{M}, \quad N = \vec{P}, \quad \frac{d\vec{P}}{dt}$$

$$\frac{d}{dt} \int_{CV} \vec{V} \rho dV + \int_{CS} \vec{V} \rho d\vec{A}$$

Macroscopic Balance – contd.

MOMENTUM

$$N = \vec{P}$$

$P \equiv$ MOMENTUM

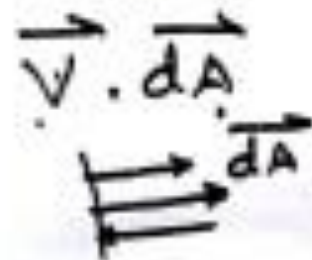
$$\vec{F} = \frac{\partial}{\partial t} \int_{CV} \vec{V} \rho dV + \int_{CS} \vec{V} \rho \vec{V} \cdot d\vec{A}$$

$$\vec{F}_S + \vec{F}_B = \frac{\partial}{\partial t} \int_{CV} \vec{V} \rho dV + \int_{CS} \vec{V} \rho \vec{V} \cdot d\vec{A}$$

└ MOMENTUM
EQN. ✓

→ i) ALL VELOCITIES ARE MEASURED
~~AND~~ RELATIVE TO THE CV

→ ii) MASS IN $\longrightarrow$ -ve
MASS OUT $\longrightarrow$ +ve.



Momentum Integral Equation

MOMENTUM INTEGRAL EQUATION

LECTURE 11.1 KOP

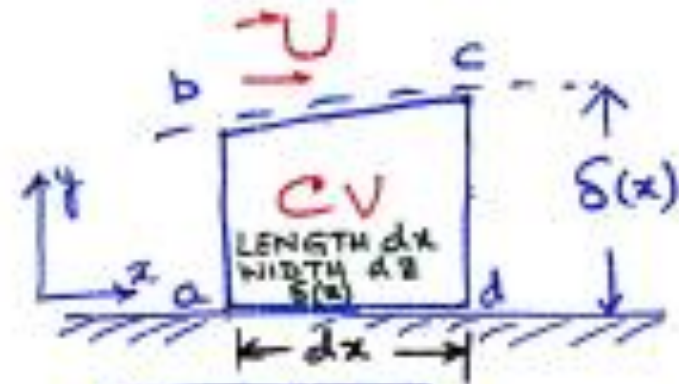
CONT. EQN $0 = \frac{\partial}{\partial t} \int_{CV} \rho dV + \int_{CS} \rho \vec{V} \cdot d\vec{A}$

STEADY, 2D FLOW.

$\int_{CS} \rho \vec{V} \cdot d\vec{A} = 0.$

$\dot{m}_{ab} + \dot{m}_{ba} + \dot{m}_{cd} + \dot{m}_{ad} = 0.$
 $\dot{m}_{ad} \approx 0.$

$\dot{m}_{ba} = -\dot{m}_{ab} - \dot{m}_{cd}.$



$\delta f''(x)$

SURF AB
LOCATED AT
x

$\dot{m}_{ab} =$

$-\int_0^{\delta} \int_0^{\delta} \rho v_x dy dz \cdot \checkmark$

$\int_A \rho \vec{V} \cdot d\vec{A}$

Momentum Integral Equation Contd.

$$\frac{cd}{x+dx}$$

$$\dot{m}_{x+dx} = \dot{m}_x + \left. \frac{\partial \dot{m}}{\partial x} \right|_x dx \quad \text{TAYLOR SER. EXPANSION.}$$

$$\dot{m}_{cd} = \left\{ \int_0^{\delta} \rho v_x dy + \frac{\partial}{\partial x} \left[\int_0^{\delta} \rho v_x dy \right] dx \right\} dz \quad \checkmark$$

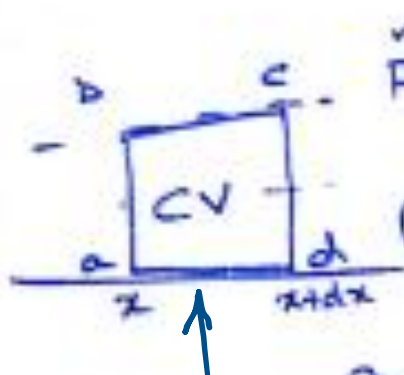
$$\dot{m}_{dc} = - \left\{ \frac{\partial}{\partial x} \left[\int_0^{\delta} \rho v_x dy \right] dx \right\} dz$$

Momentum Integral Equation Contd.

✓ COMP
OF THE
M² EQN

$$F_{Sx} + \cancel{F_{Ax}} = \cancel{\frac{\partial}{\partial t} \int_{CV} v_x \rho dV} + \int_{CS} v_x \rho \vec{V} \cdot d\vec{A}$$

$= 0$ $= 0$ SS



$$\check{F}_{Sx} = \check{m}f_{ab} + \check{m}f_{bc} + \check{m}f_{cd}$$

$\check{m}f_{bd} = 0$
Since
 $v_x = 0$
AT 'bd'

(ab) $m f_{ab} = - \left\{ \int_0^s v_x \rho v_x dy \right\} dx$

$$m f_{cd} = m f_{ab} + \frac{\partial}{\partial x} (m f_{ab}) dx$$

$$m f_{cd} = \left\{ \int_0^s v_x \rho v_x dy + \frac{\partial}{\partial x} \left[\int_0^s v_x \rho v_x dy \right] dx \right\} dx$$

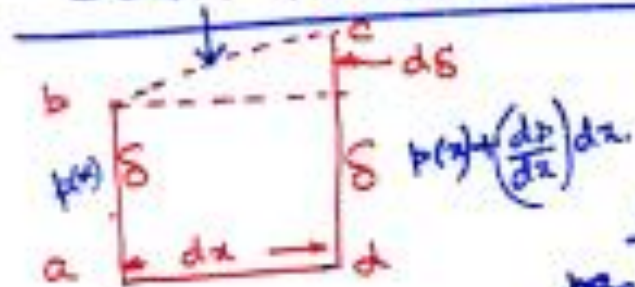
Momentum Integral Equation Contd.

$$m_{fbc} = U m_{bc}$$

$$m_{fbc} = -U \left\{ \frac{\partial}{\partial x} \left[\int_0^{\delta} p v_x dy \right] dx \right\} dz$$

$$\rightarrow \int_{cs} v_x p \vec{V} \cdot d\vec{A} = \left\{ \frac{\partial}{\partial x} \left[\int_0^{\delta} v_x p v_x dy \right] dx \right\} dz - U \frac{\partial}{\partial x} \left[\int_0^{\delta} p v_x dy \right] dz$$

SURF. FORCES ONLY IN THE x DIRECTION



F_{sx}

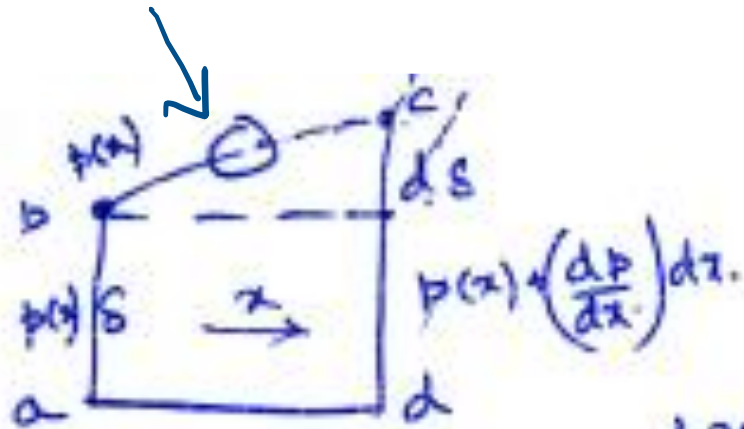
→ PRESSURE

→ SHEAR. → ∝ VEL. GRAD

$$F_{ab} = p \delta dz$$

$$F_{cd} = \left(p + \frac{dp}{dx} dx \right) (\delta + d\delta) dz$$

Momentum Integral Equation Contd.



10/10/2020

P_r on BC

$$Av. P_r = p(x) + \frac{1}{2} \frac{dp}{dx} dx$$

$$F_{bc} = \left[p(x) + \frac{1}{2} \frac{dp}{dx} dx \right] \overbrace{dS}^{AREA} dz.$$

ad

$$F_{ad} = -\tau_w dx dz$$

$\tau_w = \text{WALL SHEAR STRESS}$

Momentum Integral Equation Contd.

MT
EQN

$$\frac{\tau_w}{\rho} = \frac{d}{dx} (U^2 \Theta) + S^* U \frac{dU}{dx} \quad \checkmark$$

$$\Theta = \int_0^{\delta} \frac{v_x}{U} \left(1 - \frac{v_x}{U}\right) dy \quad S^* = \int_0^{\delta} \left(1 - \frac{v_x}{U}\right) dy$$

i) ODE

ii) WALL SHEAR STRESS $\begin{cases} \text{TURB. } \checkmark \\ \text{LAM. } \checkmark \end{cases}$

Use of Momentum Integral Equation

10 OCT
11:3 AM

USE OF MI EQ^N — ZERO PR. GRAD FLOW.

FLOW OVER A
FLAT PLATE

$$\frac{\tau_w}{\rho} = \frac{d}{dx}(U^2 \theta) + \delta^* U \frac{dU}{dx}$$

$= 0.$

→ U IS A CONST.

$$\frac{\tau_w}{\rho} = U^2 \frac{d\theta}{dx}$$

$$\frac{\tau_w}{\rho} = U^2 \frac{d\theta}{dx} = U^2 \frac{d}{dx} \int_0^{\delta} \frac{v_x}{U} \left(1 - \frac{v_x}{U}\right) dy$$

Use of Momentum Integral Equation

$$\tau_w = \rho U^2 \frac{d\delta}{dx} \int_0^1 \frac{v_x}{U} \left(1 - \frac{v_x}{U}\right) d\eta$$

$$\tau_w = \rho U^2 \frac{d\delta}{dx} \left[\int_0^1 \frac{v_x}{U} \left(1 - \frac{v_x}{U}\right) d\eta \right]$$

$\frac{y}{\delta} \equiv \eta$
 $dy = \delta d\eta$

$$\tau_w = \rho U^2 \frac{d\delta}{dx} \cdot \beta \quad \beta = \text{CONST.}$$

$$\frac{v_x}{U} = f^*(\eta)$$

$$\frac{v_x}{U} = \check{a} + \check{b}\eta + \check{c}\eta^2$$

AT $y=0$ $v_x=0$

AT $y=\delta$ $v_x=U$

AT $y=\delta$ $\frac{\partial v_x}{\partial y}=0$

$\eta=0 \quad \frac{v_x}{U}=0$
 $\eta=1 \quad \frac{v_x}{U}=1$
 $\eta=1 \quad \frac{\partial v_x / U}{d\eta}=0$

Use of Momentum Integral Equation

17-20

$$\frac{v_x}{U} = 2\eta - \eta^2 \quad \eta = y/\delta.$$
$$\frac{\tau_w}{\rho} = U^2 \frac{d\theta}{dx} = \tau_w = \rho U^2 \frac{d\delta}{dx} \left[\int_0^1 \frac{v_x}{U} \left(1 - \frac{v_x}{U}\right) d\eta \right]$$

NEWTONIAN FLUID.

$$\tau_w = \mu \left. \frac{\partial v_x}{\partial y} \right|_{y=0} = \frac{2\mu U}{\delta} = \rho U^2 \frac{d\delta}{dx} \left[\int_0^1 (2\eta - \eta^2)(1 - 2\eta + \eta^2) d\eta \right]$$
$$\frac{2\mu}{\delta \rho U} = \frac{2}{15} \frac{d\delta}{dx}$$

Use of Momentum Integral Equation

L17 WGP

$$\frac{\delta^2}{2} = \frac{15\mu}{\rho U} x + (C.)$$

AT $x=0$ $\delta=0 \Rightarrow C=0.$

$$\boxed{\frac{\delta}{x} = \frac{5.48}{\sqrt{Re_x}}}$$

BLASIUS
SOL^N $\frac{\delta}{x} = \frac{5.0}{\sqrt{Re_x}}$

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho U^2} = \frac{2\mu(U/\delta)}{\frac{1}{2}\rho U^2} = \frac{4\mu}{\rho U \delta}$$

$$C_f = \frac{0.73}{\sqrt{Re_x}}$$

$$C_f = \frac{0.664}{\sqrt{Re_x}}$$

A flat plate is installed in a water tunnel as a splitter. The plate is 0.3m long and 1m wide. The freestream speed is 2 m/s. Laminar boundary layer forms on both sides of the plate. The boundary layer velocity profile is approximated by

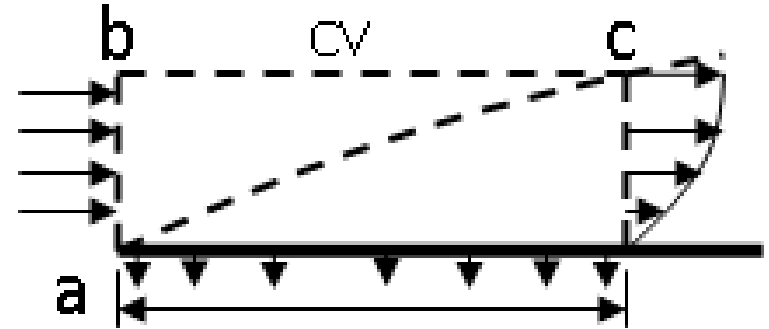
$$\frac{v_x}{U} = 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2$$

Obtain an expression for δ . Determine the total viscous drag force on the plate assuming the pressure drag is negligible ($\gamma = 1 \times 10^{-6} \text{ m}^2/\text{s}$)

Consider the steady flow of water past a porous plate with a constant suction velocity of 0.2 mm/s (i.e., $V = -0.2\mathbf{j}$ mm/s). A thin boundary layer grows over the flat plate and the velocity profile at section cd is

$$\frac{u}{U_\infty} = \frac{3}{2} \frac{y}{\delta} - 2 \left(\frac{y}{\delta} \right)^{1.5}$$

where U_{∞} is the velocity of approach at section ab and is equal to 3 m/s. Find the mass flow rate across section bc. Given: width of the plate = 1.5m, length = 2m, $\delta_{cd} = 1.5$ mm



$$\dot{m}_{ab} = -\rho \delta w U_{\infty}$$

$$\dot{m}_{bc} = ?$$

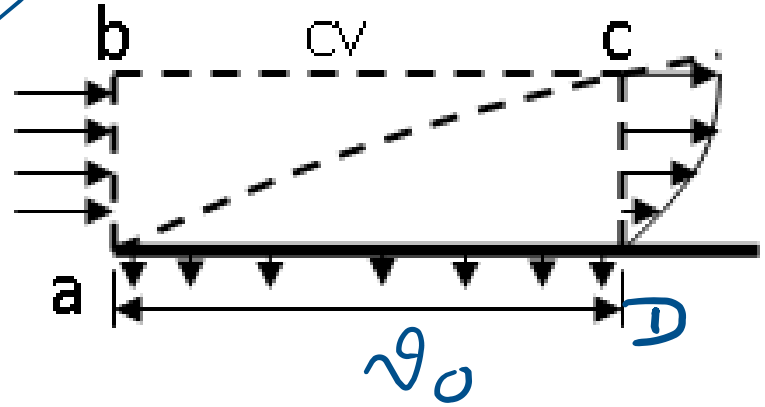
$$\dot{m}_{cd} = + \int_0^{\delta} u w dy \rho$$

$$\dot{m}_{Da} = + \rho v_0 w L$$

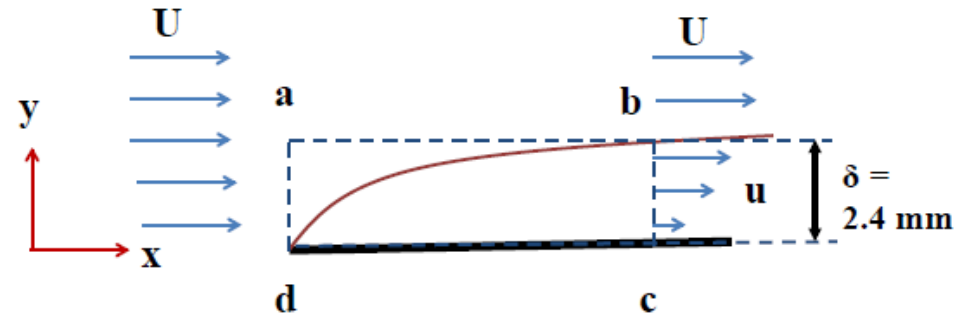
$$0 = \dot{m}_{ab} + \dot{m}_{bc} + \dot{m}_{cd} + \dot{m}_{Da}$$

$$\dot{m}_{bc} = 1042 \text{ kg/s}$$

$$\frac{u}{U_{\infty}} = \frac{3y}{2\delta} - 2\left(\frac{y}{\delta}\right)^{1.5}$$



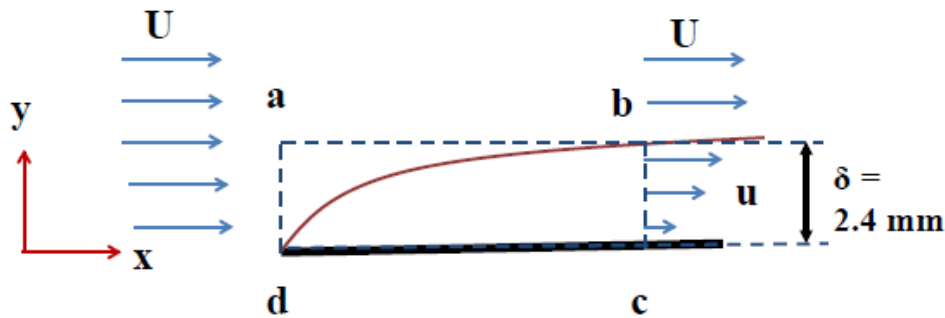
Air at standard conditions is flowing over a thin flat plate which is 1 m long and 0.3 m wide. The flow is uniform at the leading edge of the plate. The velocity profile in the boundary layer is assumed to be linear and $U = 30$ m/s. Assume that the flow conditions are independent of z and treat the flow as two dimensional. Using the control volume abcd ($dc = 1$ m), compute the mass flow rate across surface ab. The boundary layer thickness at the end of the plate (point c in the figure) is $\delta = 2.4$ mm. Determine the magnitude and direction of the x-component of the force required to hold the plate stationary ($\rho_{\text{air}} = 1.23$ kg/m³).

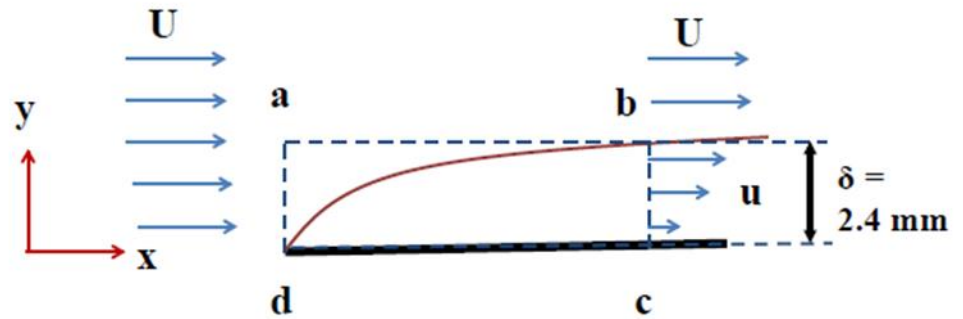


$$\dot{m}_{ab} + \dot{m}_{bc} + \dot{m}_{da} = 0$$

$$0 = \underbrace{\left\{ -\rho U b \delta \right\}}_{\dot{m}_{ad}} + \underbrace{\int_0^\delta \rho u b dy}_{\dot{m}_{bc}} + \underbrace{\dot{m}_{ab}}_{\text{circled}}$$

$$\dot{m}_{ab} = 0.0133 \text{ kg/s}$$





$$R_x \neq$$

$$mf_{ad} + mf_{ab} + mf_{bc}$$

CONVECTIVE
TIVE
m²

$$= U da \int_0^\delta \rho u b dy + \underbrace{u_{ab}}_{?} \dot{m}_{ab} + \int_0^\delta \rho u^2 b dy$$

$$= \dot{U}$$

$$\cancel{R_x} \neq \cancel{R_x} ; R_x = -0.133 \text{ N}$$

$$R_x -$$

$$R_x = \cancel{-0.133\text{ N}}$$

$R_x = \text{FORCE } \underline{\underline{\text{ON}}}$ THE CV

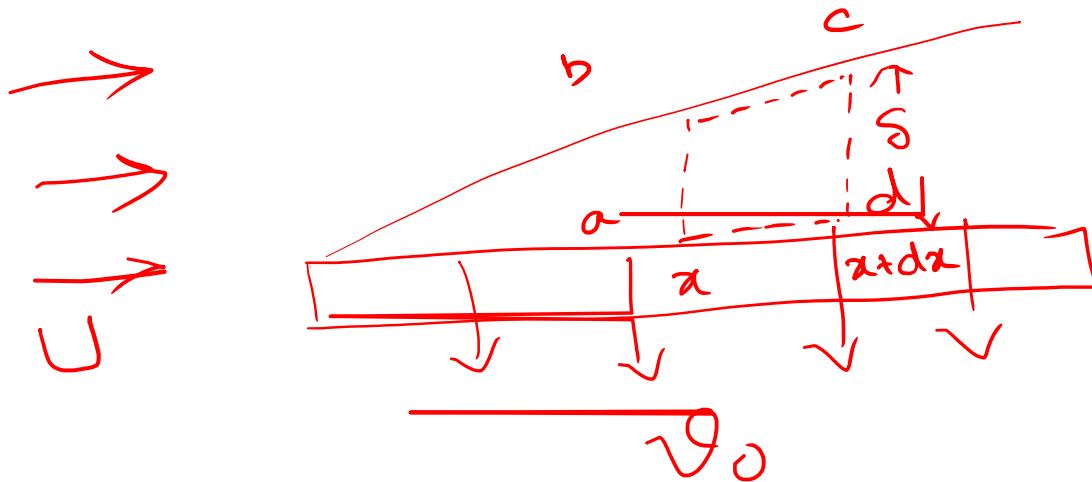
FORCE ON THE PLATE = $-R_x$
DUE TO
FLOW

~~FORCE~~ TO BE

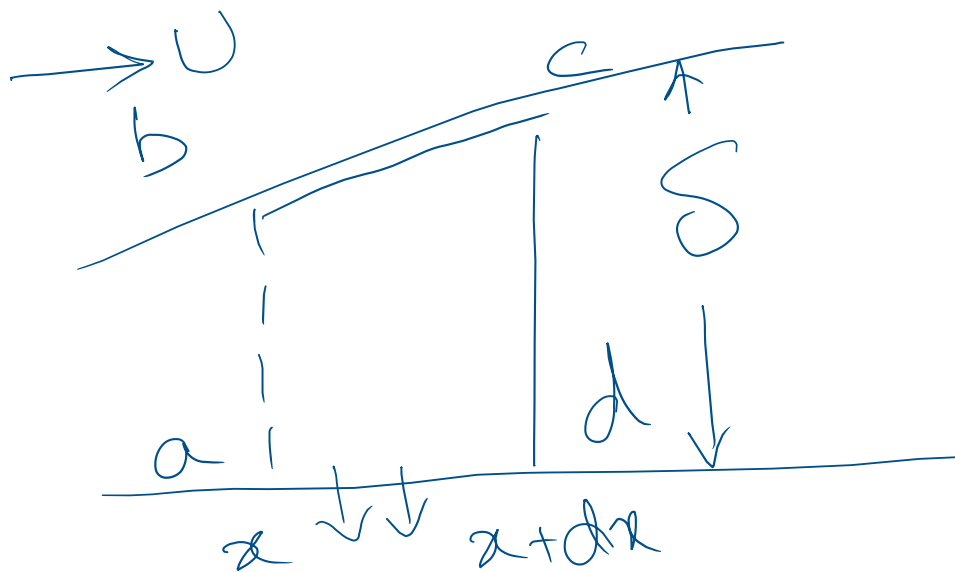
APPLIED TO
THE PLATE =
TO KEEP IT
~~FIXED~~

$$R_x \\ = -0.133\text{ N}$$

Consider horizontal, steady, incompressible flow in a boundary layer with uniformly distributed wall suction. The wall suction velocity is constant with $v = -v_0$ at $y = 0$. There is no pressure gradient. Use a differential control volume to express the axial gradient of momentum thickness ($d\theta/dx$) in terms of the wall shear stress (τ_w), v_0 , freestream velocity (U) and the density of the fluid (ρ).



$$\frac{d\theta}{dx} = f(\tau_w, v_0, U, \rho)$$



$$\Theta = \int_0^S \frac{u}{U} \left(1 - \frac{u}{U}\right) dy$$

CONT.

$$0 = \overset{\circ}{m}_{ab} + \overset{\circ}{m}_{cd} + \overset{\circ}{m}_{bc} + \overset{\circ}{m}_{da}$$

$$\overset{\circ}{m}_{cd} = \overset{\circ}{m}_{ab} + \frac{d}{dx} (\overset{\circ}{m}_{ab}) \Delta x$$

$$\dot{m}_{ad} = - \int_0^{\delta} \rho u w dy$$

$$\dot{m}_{cd} = \text{TAYLOR SER.}$$

$$\dot{m}_{ad} = + \rho v_0 w L$$

$$\dot{m}_{bc} = ?$$

$$\dot{m}_{bc} =$$

✓ Expts.

m² EQN

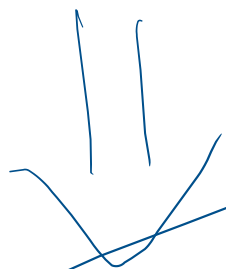
$$\dot{m}_{ab} = - \int_0^S \rho U^2 w dy$$

$$\dot{m}_{cd} = \text{TAYLOR SER} \quad \dot{m}_{ab}$$

$$\dot{m}_{bc} = \dot{m}_{bc} \times U$$

$$m_{fad} = \int w \quad w dx$$

$$T_w = ?$$



$$\frac{d\theta}{dx} = \frac{T_w}{\rho U^2} - \frac{v_0}{U}$$